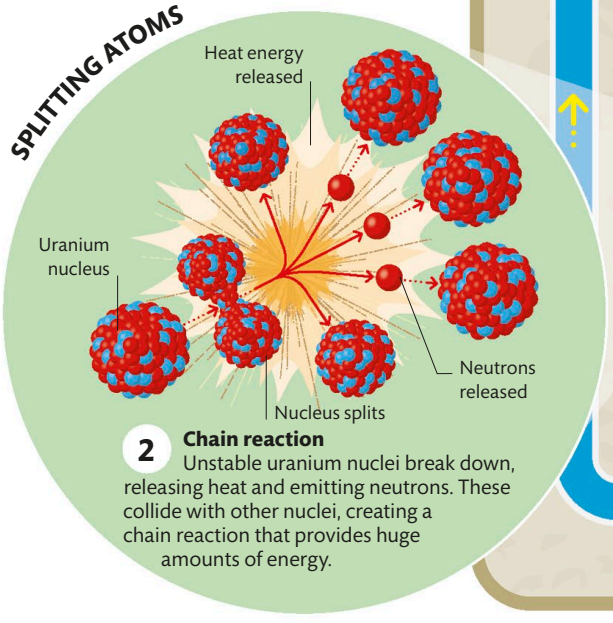
Nuclear power

Nuclear energy is released when the nuclei of atoms are either split apart (nuclear fission) or fused together (nuclear fusion). A nuclear power plant harnesses the energy released from fission to generate electricity.

Nuclear fission

Nuclear power plants are fueled by radioactive elements such as uranium. When atoms of the fuel are split, a huge amount of energy is released as heat. This heat turns steam-driven turbines, which power electricity generators. Nuclear fission uses small amounts of fuel and produces far less greenhouse gas emissions than fossil fuels.



2 Chain reaction
Unstable uranium nuclei break down, releasing heat and emitting neutrons. These collide with other nuclei, creating a chain reaction that provides huge amounts of energy.

Inside a nuclear reactor
Nuclear fission takes place in a reactor encased in a strong, reinforced concrete dome designed to contain the radiation emitted.

4 Creating steam
Heated by the reactor core, the water flows into a heat exchanger, giving up its energy to a second closed system of piping that carries cold water. The cold water is turned into hot steam, under high pressure.

3 Control rods
Control rods modify the speed of the chain reaction. When lowered in among the fuel rods, they absorb many of the free neutrons to slow down the reaction.

Raising control rods speeds up reaction

CONTROL RODS

HEAT EXCHANGER

REACTOR VESSEL

FUEL RODS

Control rods lowered to soak up neutrons and slow down chain reaction

Water in reactor core

1 Uranium fuel rods
Hundreds of metal rods grouped into bundles contain small pellets of uranium fuel. They are lowered into the reactor's core.

REACTOR CORE

Pump maintains flow of water

PUMP